Willow WU

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Data / AI Infrastructure Engineer — Python/SQL · DuckDB/Parquet · ETL Pipelines · Model Serving / HPC

EDUCATION

Columbia University *New York, NY*
M.S. in Computer Science *(Machine Learning Track)* — GPA 3.96/4.0 *Sep 2025 – Dec 2026*
Relevant Coursework: Analysis of Algorithms, Applied Machine Learning, Deep Learning, Natural Language Processing, Engineering Software-as-a-Service (SaaS), Advanced Spoken Language Processing

Hong Kong Baptist University (HKBU) *Hong Kong*
B.Sc. in Data Science *Sep 2021 – Jun 2025*

EXPERIENCE

Columbia Business School — Research Assistant *New York, NY*
Prof. Emmanuel Yimfor — VC/PE Founder-Diversity Research *Jun 2026 – Present*

- Engineered a scalable Python/DuckDB feature pipeline over a **962,757-row** roster and **543,016** image-labeled profiles, generating predictions for **419,741** no-photo profiles on CBS GRID HPC.
- Used raw-SQL joins, Parquet intermediates, and SGE batch execution; replaced a costly full nested-Parquet merge with a targeted full-coverage feature join, cutting compute and data movement.
- Built validation tooling for spot-check sampling, class-distribution comparison, and photo-vs-no-photo drift analysis; improved macro-F1 **0.71** → **0.79** on 5-fold OOF.

Microsoft — Software Engineer Intern *Summer 2026*
OfficePLUS AI Assistant

- Diagnosed production LLM-assistant failures across **3 non-joinable telemetry stores** (Azure SQL + 2 App Insights tenants) with KQL and T-SQL over **11.5K+** failure events.
- Proved **97.6%** of "rate-limit" failures were genuine quota exhaustion vs 2.4% throttle (11,958 vs 295), correcting the team's hypothesis and falling back from `messageId` to `conversationId`+time for cross-store joins.
- Built a routing-verification probe proving **500/500** concurrent runs routed correctly with **zero** cross-session cross-talk, complementing telemetry analysis with direct systems validation.

China Academy of Railway Sciences — Data Engineering Intern *Beijing, China*
Institute of Electronic Computing Technology *Summer 2023*

- Cleaned, migrated, and validated **3M+** freight records on a DCFS/ELK/PCF Spring Cloud data platform; designed APIs for data retrieval and behavior tracking across distributed systems.

PUBLICATIONS & PROJECTS

Explainable Dynamic GNNs for Predictive Maintenance in Vehicle Chassis Systems *First Author* *2025*
Proceedings of Machine Learning Research (PMLR), **ACML 2025**.

- Processed ∼**9M** noisy fault-log records across ∼**100GB** of scattered industrial data; identified source misalignment and reframed the task around maintenance-linked labels.
- Constructed weighted fault graphs (temporal, spatial, Apriori edges); evaluated with vehicle-wise chronological CV to prevent leakage.

ML Inference Service — FastAPI, Docker, batch inference *Personal* *2026*

- Built a Dockerized FastAPI inference service with single-record and batch scoring, health checks, and a swappable model interface; benchmarked ∼**17 ms** p50 / ∼**30 ms** p95 latency and ∼**78K records/s** batch throughput in local CPU tests.

SKILLS

Languages: Python, SQL, TypeScript, C++
Data / Pipelines: ETL, Data Pipelines, DuckDB, Parquet, pandas, NumPy, Feature Engineering, Batch Processing
Infra / Serving: FastAPI, Docker, Model Serving, Health Checks, Batch Inference, Distributed Systems, HPC (SGE/GRID)
Platforms: Azure (App Insights/KQL, Azure SQL, DevOps), Spring Cloud, Elasticsearch/ELK, PostgreSQL, SQLite

HONORS & AWARDS

Kaggle Silver Medal (Top 4.2%, 2024) • **MCM Honorable Mention** (Top 20%, 2024) • **CUMCM Provincial 2nd Prize** (Top 16%, 2023) • Merit Scholarship (Top 15%)